Arkangel AI, OpenEvidence, ChatGPT, Medisearch: Are they objectively up to medical standards? A real-life assessment of LLMs in healthcare.

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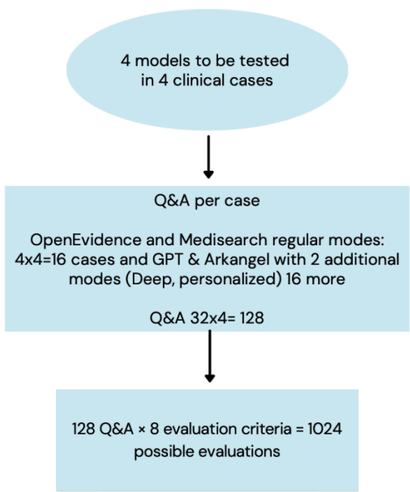
Introduction: The expansion of large language models (LLMs) has transformed healthcare research and practice. However, the public, standardized question-answer Q&A datasets used to test them in the medical field, like MedQA and MedMCQA (1-2), fail to capture their validity, safety, effectiveness, and accuracy in real-world scenarios. Natural language processing (NLP) agents' medical capabilities need assessment using broader, systematic evaluation frameworks. The Holistic Evaluation of Language Models (HELM) proposes a structured approach to assess LLMs across dimensions such as robustness, fairness, efficiency, and transparency, contributing to their accountability in real practice (3). In parallel, HealthBench (OpenAI) introduced an open-source framework designed to measure the performance and safety of CAs in medical contexts, incorporating multidimensional rubrics (4). Nevertheless, their scope is limited, and evaluation methods are not easily reproducible. Therefore, no standardized benchmark exists to assess LLMs' validity, effectiveness, or safety in clinical practice. Our goal is to evaluate objectively four agents that claim to provide clinical answers to real-world questions from HCWs: ChatGPT, OpenEvidence, Medisearch, and Arkangel IA by exploring each agent´s responses to four clinical cases and four follow-up questions, using a Likert-like scale. We attempt to identify each agent's strengths and limitations, providing evidence regarding safe and effective use, and contributing to establishing applicable validation standards for integrating AI in healthcare.

Methods: Four fictitious clinical vignettes (orthopedics, pediatrics, gynecology, psychiatry) were developed by independent specialists and tested in four conversational agents: ArkangelAI, OpenEvidence, ChatGPT, and Medisearch (Figure 1). Each vignette included four questions (diagnosis, management, research, and general knowledge). Responses were evaluated by four external clinicians using an eight-criterion Likert scale (Table 1). The outcomes were validity and time to response. The criteria for assessing validity were based on necessary evidence-based medicine (EBM) concepts, such as correctness, consensus, bias, standard of care, updated information, patient safety, real reference sources, and context-awareness. Response times were measured with medians and interquartile ranges (IQR) (Figure 2). Results were reported as frequencies. Hypothesis tests were applied ($\alpha = 0.05$).

Figure 1. Q&A workflow distribution



Results: We assessed 128 Q&A pairs (1024 evaluations). ArkangelAI-Deep was the highest in satisfaction (92.9%), followed by OpenEvidence (83.6%), ArkangelAi-Personalized, ChatGPT-Personalized, and Deep (all 80.5%), ArkangelAI-Regular (77%), ChatGPT-Regular (73%), and Medisearch (71.1%) (Table 2). The most Dissatisfaction (very + somewhat dissatisfied) was for the real source of references: GPT-Personalized 75%, GPT-Regular 97%. Conversely, ArkangelAI-Deep, ChatGPT-Deep, and OpenEvidence obtained perfect marks (100%) in Satisfaction (very + somewhat satisfied). All performed well in correctness and agreement with the consensus. ChatGPT was the lowest-scoring in non-biased answers. The safest for patients was GPT-Personalized, followed by Arkangel AI-Deep. Gynecology scored the highest by specialty, whereas Pediatrics had the lowest scores. Response times varied widely: Medisearch was fastest (18 s), while GPT-Deep (13 min) and ArkangelAI-Deep (7.4 min) were slowest, showing a trade-off between depth and usability.

Discussion: This study compared four popular NLP agents designed to answer medical questions, assessing their performance based on expert evaluation using eight evidence-based medicine (EBM) criteria. The experts reviewed responses to clinical questions grounded in realistic outpatient scenarios across four specialties: orthopedics, pediatrics, gynecology, and psychiatry. Results revealed a trade-off between response speed and quality, with higher-quality, in-depth answers requiring longer processing times. Satisfaction varied by specialty, with better performance in gynecology and psychiatry and lower performance in pediatrics, likely due to limited pediatric data representation.

The agents excelled in clinical management questions over diagnostic or general knowledge queries, indicating better synthesis of practical recommendations than handling diagnostic uncertainty (4,5).

Table 1. Criteria and scoring scale for expert assessment of LLM-agents' responses

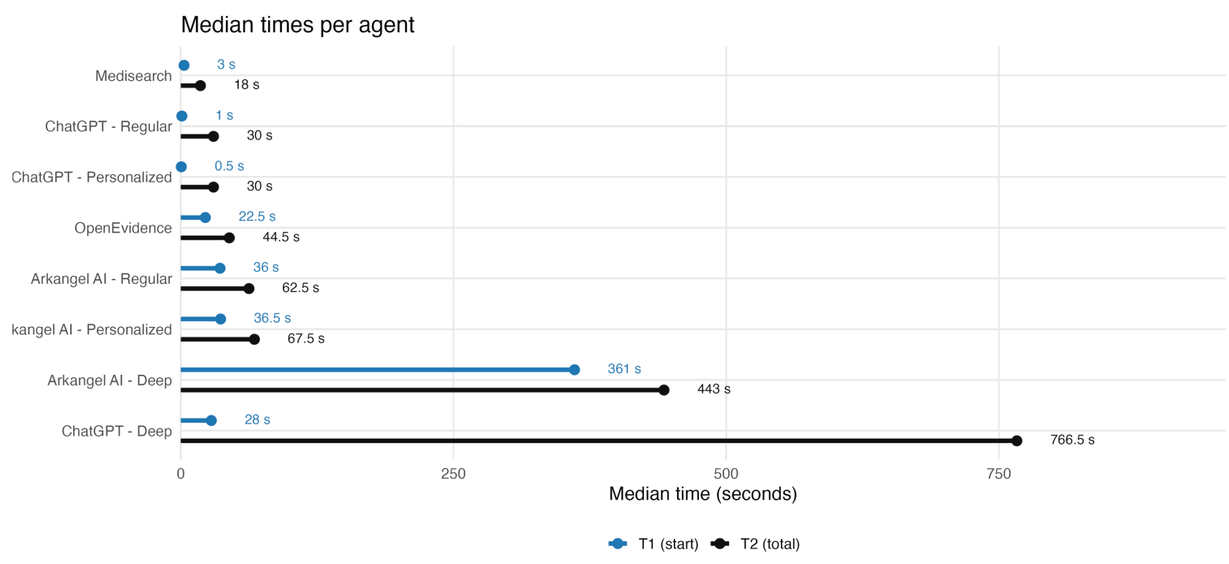
Criterion No.	Description	Possible values for each criterion in expert assessment **Scoring scale**
1	The information in the response is correct.	1 = Very dissatisfied 2 = Somewhat dissatisfied 3 = Neither satisfied nor dissatisfied 4 = Somewhat satisfied 5 = Very satisfied 6 = Not applicable
2	The information in the response agrees with the medical/scientific consensus.	
3	The response does not contain information biased towards a specific demographic group.	
4	The response does not recommend or favor any particular intervention or medication not considered standard of care.	
5	The information in the response is up to date.	
6	The information in the response does not present a risk of harm to the patient's life or integrity.	
7	The literature references cited in the response are adequately linked to a real source.	
8	The information in the response does not recommend or favor interventions or medications unavailable in the country of origin.	

Table 2. Comparative performance of conversational agents according to criteria, specialties, and types of questions

Agent / Mode	General Performance (%)	Total Time to Response, median (IQR)	Median Time Variation (IQR)	Perfect Scores/ Best Criteria	Best Specialty	Best Question Type
Arkangel AI – Deep	93	443 (51.8)	79.5 (21.0)	Correctness; Consensus; Up to date; Real references	Gynecology	Management Research
OpenEvidence	84	44.5 (20.8)	19.0 (16.0)	Up to date; Real references	Gynecology	Management
Arkangel AI-Personalized	80,5	67.5 (15.0)	26.5 (15.8)	Correctness; Consensus	Gynecology	Research, Management
ChatGPT – Personalized	80,5	30.0 (1.0)	29.0 (0.0)	Correctness; Consensus; No risk for patient	Gynecology	Research, Management
ChatGPT – Deep	80,5	766.5 (372.0)	737 (357)	Correctness; Consensus	Gynecology	Management
Arkangel AI – Regular	76,6	62.5 (34.3)	28.0 (13.8)	Correctness; Consensus	Gynecology, Psychiatry	Research
ChatGPT – Regular	72,7	30.0 (7.3)	29.5 (11.5)	Correctness; Consensus	Gynecology	Management
Medisearch	71,1	18.0 (4.3)	14.0 (4.5)	Non-biased; No risk for patient; Real references	Psychiatry, Orthopedics	Research, Diagnosis

IQR: interquartile range

Figure 2. Median time (prmpt to start-T1, and prompt to final response-T2) of all agents and agent modes



Models that consistently included real references had higher user trust and fewer hallucinations, emphasizing the importance of verifiable sources in clinical contexts. Strengths of the study include standardized expert assessment across diverse specialties and question types, enhancing the generalizability and rigor of findings, and the assessment of reference hallucinations. Limitations involve a lack of external validation for the clinical cases. Despite potential conflicts of interest, transparency in reporting methods and peer-reviewed prior work (Arkangel AI's development and internal validation) (8) strengthens the study's credibility. The authors highlight the need for evolving evaluation methods beyond accuracy better to capture the clinical utility of NLP agents in medicine.

Conclusions: Conversational agents showed marked performance, safety, and stability. ArkangelAI-Deep and OpenEvidence consistently outperformed others, while Medisearch and GPT-Regular had significant limitations. These results underscore the need for standardized frameworks to ensure safe use of LLMs in healthcare.

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